ODE HW 4.6

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problem setup:

$$E(t) = 12\cos(\omega t).$$

part (a):

homogeneous solution

the homogeneous equation $I'' + 4I = 0$ has roots $r = \pm 2i$, giving

$$I_h = A\cos(2t) + B\sin(2t).$$

particular solution

since the forcing function is $-12\omega\sin(\omega t)$ and $\omega \neq 2$, we guess

$$I_p = D\sin(\omega t).$$

substituting:

$$-D\omega^2\sin(\omega t) + 4D\sin(\omega t) = -12\omega\sin(\omega t)$$
$$D(4-\omega^2) = -12\omega \implies D = \frac{12\omega}{\omega^2 - 4}.$$

so

$$I_p = \frac{12\omega}{\omega^2 - 4}\sin(\omega t).$$

apply initial conditions

general solution is: $I = I_h + I_p$:

$$I(t) = A\cos(2t) + B\sin(2t) + \frac{12\omega}{\omega^2 - 4}\sin(\omega t).$$

from $I(0) = 0$:

$$A = 0.$$

from $Q(0) = 0$, gives $I'(0) = E(0)/L = 12$:

$$I'(0) = 2B + \frac{12\omega^2}{\omega^2 - 4} = 12 \implies B = \frac{12 - \frac{12\omega^2}{\omega^2-4}}{2} = \frac{-12}{\omega^2 - 4}.$$

solution

$$\boxed{I(t) = \frac{12}{\omega^2 - 4}\Big(\omega \sin(\omega t) - 2\sin(2t)\Big)}$$

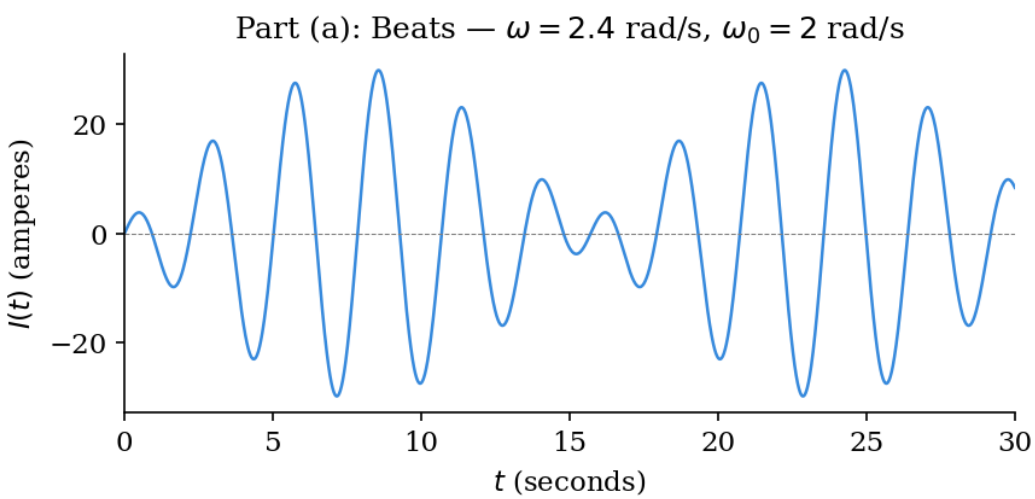

Figure 1: current $I(t)$ for $\omega = 2.4$ rad/s. (made with GraphPad)

ppart (b):

$$I_p = t\big(A\cos(2t) + B\sin(2t)\big).$$

computing I_p'' and substituting into $I'' + 4I = -24\sin(2t)$:

$$-4A\sin(2t) + 4B\cos(2t) = -24\sin(2t) \implies A = 6, \quad B = 0.$$

so $I_p = 6t\cos(2t)$.

general solution is:

$$I(t) = A\cos(2t) + B\sin(2t) + 6t\cos(2t).$$

$I(0) = 0$ gives $A = 0$. $I'(0) = 12$ gives $B = 0$.

solution

$$\boxed{I(t) = 6t\cos(2t)}$$

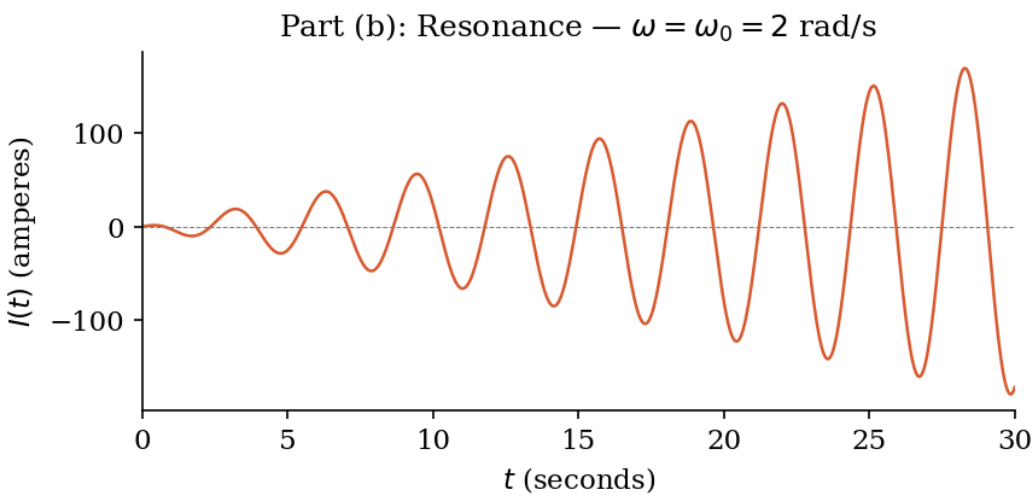

Figure 2: current $I(t) = 6t\cos(2t)$ at resonance. (made with GraphPad)